

42115 Network Optimization

Flowing containers in a liner shipping network

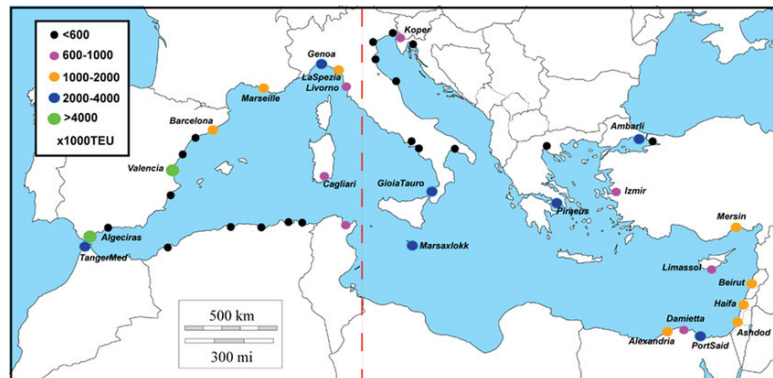
Project Assignment, David Pisinger, DTU 2022

(new) In this project assignment we will see how cargo flowing in a liner shipping network can be modeled and solved using various network optimization models.

The assignment consists of four independent parts: In the first part we study how a customer can find the cheapest way of transporting his containers through a liner shipping network. In the next part we study the problem of repositioning empty containers. Then, we study how an weighted multicommodity flow problem can be solved through column generation. Finally, we study various ways of modeling a hub port in a multicommodity flow problem.

The project should be solved in **groups of 2-3 people**. The report may be written in Danish or English. The report should be handed in as *one* pdf-file to DTU-inside no later than **Sunday 4th December at 8:00**. It is allowed to write by hand (e.g. drawing the figures) and scan the pages on a good scanner. It is not allowed to hand-in the project alone (group with 1 person), except if you get a written permission from the course responsible before 11th November.

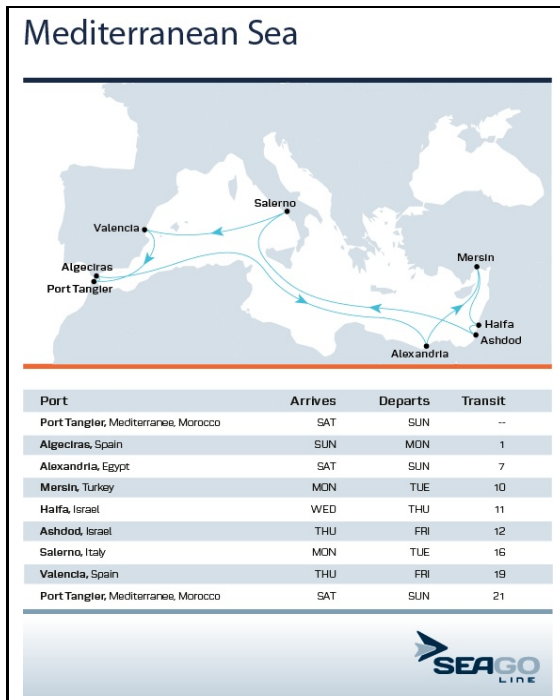
Please read the whole text including the hints before starting to solve the assignment.



Introduction

Containerized transportation is one of the most important inventions in logistics during the last century, giving us access to cheap and fast transportation of goods from all around the world. The new triple-E vessels can carry more than 18.000 containers, resulting in low freight rates and a small carbon footprint per container.

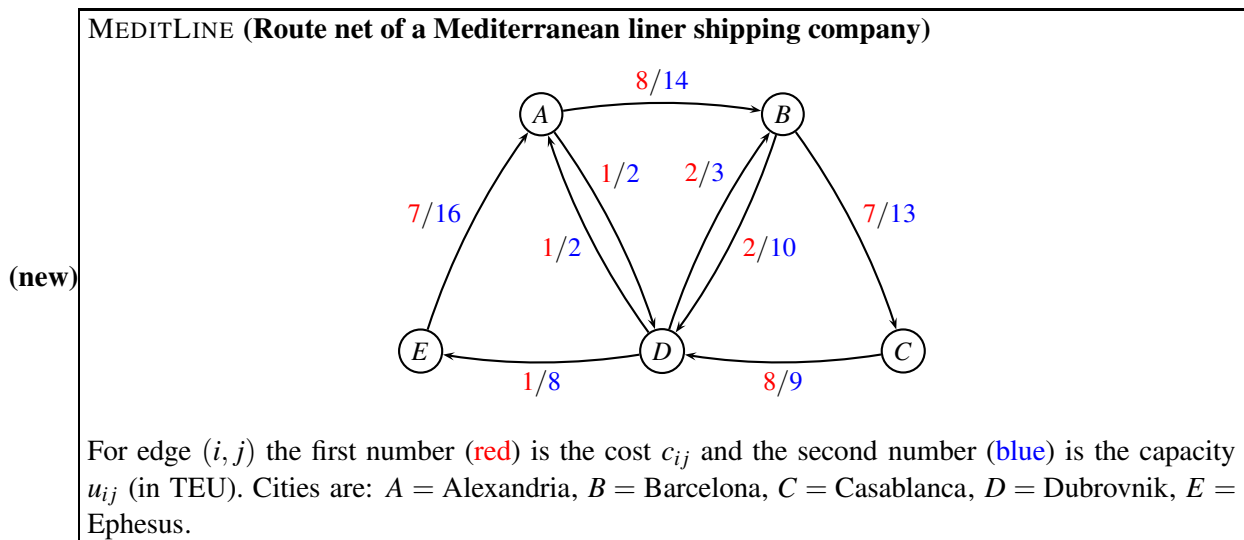
The vessels sail according to a route plan (like busses) as illustrated in the figures below. Weekly departure in each port is ensured by deploying additional vessels if the round trip takes more than 7 days. As an example, the Mediterranean roundtrip takes 21 days, so three vessels are deployed to this route. In a similar way, two vessels are assigned to the Ligurian Sea roundtrip as it takes 14 days.



When goods are transported from one customer to another, it may be necessary to transship goods at a port. *Transshipment* means unloading the container from one vessel and storing it on the quay before it can be loaded to the next vessel. For instance, a container from Alexandria to Barcelona will follow the Mediterranean line to Valencia, where it is transshipped to the Ligurian line in order to be transported to the final destination.

The below instance, **MEDITLINE**, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost (per TEU container) $c_{ij} \geq 0$, and a capacity $u_{ij} \geq 0$.



Containers appear in various sizes. The most common sizes are the 20 foot container, the 40 foot container and the 40 foot high-cube container.



The capacity of a vessel is measured in TEU (*Twenty-foot Equivalent Units*). A 20 foot container takes up 1 TEU space, a 40 foot container takes up 2 TEU space, while a 40 foot high-cube container takes up 4 TEU space. For obvious reasons the 2 TEU space occupied by a 40 foot container, and the 4 TEU space occupied by a high-cube container, may not be split and have to follow the same route from source to destination.

Finding the cheapest route through a network

Each customer is interested in finding the cheapest way of transporting his containers from a source s to a destination t . Assuming that the customer wants to send a container of size w TEU, we must ensure that all the w TEU follow the same path, since the container is *unsplittable*.

Question 1 (2 point) Given a network $G = (V, E, c, u)$, with costs $c \geq 0$ (per TEU), and capacities $u \geq 0$ (in TEU). Describe how you can modify the graph, such that the cheapest route from s to t for one container of size w TEU can be found using the Bellman-Ford algorithm. The output of the algorithm should be the cost of routing w TEU. ■

Question 2 (3 point) Implement the algorithm in an programming language (e.g. Julia, C++, Java) without using a MIP model. The algorithm should output the path and the cost of routing a container of size w TEU. ■

The algorithm should be able to read a file in a format

```
n m s t w
a1 b1 c1 u1
a2 b2 c2 u2
: : : :
am bm cm um
```

The first line states the number n of nodes, number m of edges, source s , destination t , and size w . The following lines specify for each edge (a_i, b_i) the cost (per TEU) c_i and capacity u_i (in TEU).

Question 3 (1 point) Run the code for MEDITLINE using source $s = E$, destination $t = D$ assuming that a 40 foot high-cube container is transported (i.e. $w = 4$). Report the solution (cost and path) in the report. ■

Question 4 (2 point) Now assume that the found route may visit at most h nodes (including the source and destination). How would you solve the problem as a network optimization problem (not using a MIP)? You do not need to implement the code. ■

Finding bottlenecks in the network

The shipping company is interested in finding an upper limit on the how much can be transported from a region S to another region T .

Question 5 (2 point) Given a network $G = (V, E, c, u)$, with costs $c \geq 0$ (per TEU), and capacities $u \geq 0$ (in TEU). Argue which network algorithm that can be used to find the the largest amount of containers that can be sent from region S to T . We assume that all containers have size 1 TEU. ■

Question 6 (3 point) For the MEDITLINE instance show how you find the capacity from region $S = \{A, D\}$ to region $T = \{B, C, E\}$. Give sufficient details to follow the calculations. You should use a network algorithm not a MIP-solver. ■

The shipping company also wants to estimate the security of supply. Two paths between ports s and t are *edge disjoint* if they do not share any edge. The more edge disjoint paths we have, the higher security of supply.

Question 7 (2 point) Given a network $G = (V, E, c, u)$, with costs $c \geq 0$ (per TEU), and capacities $u \geq 0$ (in TEU). Describe how you can find the maximum number of edge disjoint paths between ports s and t as a network problem. ■

Question 8 (3 point) For the MEDITLINE instance calculate the maximum number of edge disjoint paths from port $s = B$ to port $t = A$. Give sufficient details to follow the calculations. You should use a network algorithm not a MIP-solver. ■

Repositioning of empty containers



Due to the imbalance in trade, empty containers tend to accumulate at various ports. For instance China is shipping much more goods to Europe than the other way, resulting in a surplus of empty containers in Europe and a demand of empty containers in China.

(new) For MEDITLINE, one could have the following surplus (in TEU) of containers:

node i	A	B	C	D	E
surplus σ_i	7	-6	-5	0	4
demand d_i	-7	6	5	0	-4

It is assumed that all empty containers are equal and have size 1 TEU.

Question 9 (2 point) Given a network $G = (V, E, c, u)$, and a set of surpluses σ_i for each node $i \in V$, formulate the repositioning of empty containers as a known network optimization problem. ■

Question 10 (5 point) Solve (by hand) the MEDITLINE repositioning problem, using a network algorithm (not CPLEX). State which algorithm you use, and give sufficient details to make it possible to follow your calculations. Report the optimal solution. ■

Flowing containers

Frequently there are several different ways of transporting containers, hence the task of the shipping company is to utilize the network in the best possible way, respecting the overall capacity of the container vessels. Large containers (40 foot container of size $w = 2$ or high-cube of size $w = 4$) take up several TEU space but they cannot be split along a route. Hence we need to make sure that all the TEU are sent along the same path. We can formulate this problem as a variant of the multi-commodity flow problem.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost $c_{ij} \geq 0$ (per TEU) and a capacity $u_{ij} \geq 0$ (in TEU). Moreover we are given a number of commodities $K = \{1, \dots, m\}$, each defined by the values (s_k, t_k, d_k, w_k) , where s_k denotes the source of commodity k , t_k denotes the target of commodity k , $d_k > 0$ denotes how many containers to be sent, and $w_k > 0$ denotes how much space (in TEU) each container takes up.

Using integer decision variables x_{ij}^k to indicate how many containers of commodity $k \in K$ are using edge $(i, j) \in E$, we get the following formulation of the *weighted multicommodity flow problem*

$$\min \sum_{k \in K} \sum_{(i,j) \in E} c_{ij} w_k x_{ij}^k \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = 0 \quad k \in K, j \in V \setminus \{s_k, t_k\} \quad (2)$$

$$\sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = -d_k \quad k \in K, j = s_k \quad (3)$$

$$\sum_{i \in V} x_{ij}^k - \sum_{i \in V} x_{ji}^k = d_k \quad k \in K, j = t_k \quad (4)$$

$$\sum_{k \in K} w_k x_{ij}^k \leq u_{ij} \quad (i, j) \in E \quad (5)$$

$$x_{ij}^k \in \mathbb{N} \quad k \in K, (i, j) \in E \quad (6)$$

The objective function (1) minimizes the cost of flowing all containers. The first constraints (2) ensure flow conservation. The next constraints (3) ensure for each k that the commodity flows out from s_k . The following constraints (4) ensure for each k that the commodity flows into t_k . Finally, the capacity of each edge needs to be respected by (5). In (6) we have $\mathbb{N} = \{0, 1, 2, \dots\}$, hence the domain of the variables ensure that containers are not split. We will call this formulation the *edge formulation* of the *weighted multicommodity flow problem*.

The weighted multicommodity flow problem is $\mathcal{NP}$ -hard, as shown by Khan (2000), meaning that the problem may be very difficult to solve for large-sized instances.

We consider again the route net MEDITLINE where we now have the demands

k	s_k	t_k	d_k	w_k
1	A	D	3	1
2	E	D	4	2
3	B	A	2	4

Commodity $k = 1$ is shipping three 20 foot containers ($w_k = 1$) from A to D , commodity $k = 2$ is shipping four 40 foot containers ($w_k = 2$) from E to D , and commodity $k = 3$ is shipping two 40 foot high-cube containers ($w_k = 4$) from B to A .

Question 11 (3 point) Write up the edge formulation for MEDITLINE and solve it as an IP-problem by use of CPLEX. Enclose the CPLEX model and CPLEX solution. Report the selected route of each container in a table. ■

Question 12 (1 point) Solve the edge formulation for MEDITLINE as an LP-problem. Enclose the CPLEX model and CPLEX solution. Report the selected route of each container in a table. ■

Question 13 (1 point) The LP-solution gives a lower bound for the IP problem. Compare the IP and LP solution, and discuss whether the LP-relaxation provides a good lower bound. ■

Question 14 (3 point) For each commodity k in MEDITLINE, write up a table with all possible simple routes from s_k to t_k through the graph $G = (V, E, c, u)$. A route may only use edges where the capacity is at least w_k TEU, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the w_k TEU. (Hint: There should be 8 distinct routes.) ■

Let R_k denote the set of possible routes from s_k to t_k allowing w_k TEU of capacity. Let moreover $a_{r,ij}^k \in \{0,1\}$ denote if route r for commodity k is using edge (i,j) , and let $\hat{c}_r^k$ denote the cost for sending w_k containers along route $r \in R_k$. Let x_r^k for each route $r \in R_k$ denote how many containers are using this route for commodity k . We can now write up an alternative formulation of the problem, which we will call the *path formulation of the weighted multicommodity flow problem*.

$$\min \sum_{k \in K} \sum_{r \in R_k} \hat{c}_r^k x_r^k \quad (7)$$

$$\text{s.t.} \sum_{k \in K} \sum_{r \in R_k} w_k a_{r,ij}^k x_r^k \leq u_{ij} \quad (i,j) \in E \quad (8)$$

$$\sum_{r \in R_k} x_r^k \geq d_k \quad k \in K \quad (9)$$

$$x_r^k \in \mathbb{N} \quad k \in K, r \in R_k \quad (10)$$

The objective function (7) minimizes the cost of the selected routes. Constraint (8) ensures that the capacity of each edge is respected, while (9) ensures that the demanded amount of goods are sent from s_k to t_k . Finally, (10) ensures that containers are not split in parts. We use greater-equal signs in (9), as they are easier to handle for the LP-solver, but an optimal solution will never use more than one path for each commodity.

Question 15 (3 point) Write up the path formulation of MEDITLINE. You can use CPLEX format. ■

Question 16 (1 point) Solve the path formulation of MEDITLINE to IP-optimality with CPLEX and report the lower bound. Enclose the model. ■

Question 18 (1 point) Compare the IP and LP solution, and discuss whether the LP-relaxation provides a good lower bound. ■

Question 19 (5 point) (*Theoretical question*) Prove that (in general) you will find tighter bounds by solving the LP-relaxation of the path formulation than solving the LP-relaxation of the edge formulation. (Hint: you need to show that any feasible solution to the LP-relaxed path formulation also is a feasible solution to the LP-relaxed edge formulation. Moreover you should show that at least one instance exists for which the path formulation gives tighter bounds than the edge formulation. You may assume that (9) is satisfied with equality constraints.) ■

A disadvantage of the path formulation is that many routes from s_k to t_k may exist for each commodity, such that the path formulation in practice becomes unsolvable due to the large number of decision variables.

If we only are interested in solving the LP-relaxed problem, this problem can be avoided by writing up the routes only when they are needed in the simplex algorithm, hoping that just a small subset of the routes will be generated. This approach is known as *column generation*.

(new) To illustrate the principle, consider again the LP-relaxed path formulation of MEDITLINE, where we initially have found a greedy solution using the following paths:

commodity k	route	cost
1	ABCD	23
1	ABD	10
2	EABD	34
3	BCDEA	92

This leads to the following LP-model corresponding to (7) - (10):

$$\begin{array}{llllllll}
\min & 23x_1^1 & + & 10x_2^1 & + & 34x_1^2 & + & 92x_1^3 \\
\text{s.t.} & x_1^1 & + & x_2^1 & + & 2x_1^2 & & \leq 14 & (y_{AB}) \\
& & & & & & & \leq 2 & (y_{AD}) \\
& x_1^1 & & & & + & 4x_1^3 & \leq 13 & (y_{BC}) \\
& & & x_2^1 & + & 2x_1^2 & & \leq 10 & (y_{BD}) \\
& x_1^1 & & & & + & 4x_1^3 & \leq 9 & (y_{CD}) \\
& & & & & & & \leq 2 & (y_{DA}) \\
& & & & & & & \leq 3 & (y_{DB}) \\
& & & & & & 4x_1^3 & \leq 8 & (y_{DE}) \\
& & & & & 2x_1^2 & + & 4x_1^3 & \leq 16 & (y_{EA}) \\
& x_1^1 & + & x_2^1 & & & & \geq 3 & (\bar{y}_1) \\
& & & & & x_1^2 & & \geq 4 & (\bar{y}_2) \\
& & & & & & x_1^3 & \geq 2 & (\bar{y}_3) \\
& x_1^1, x_2^1, x_1^2, x_1^3 \geq 0
\end{array} \tag{11}$$

We call the above model, the *restricted master problem*.

Let y_{ij} denote the dual variable corresponding to constraint (8) for edge (i, j) , and let $\bar{y}_k$ denote the dual variable corresponding to constraint (9) for commodity k . The dual variables can be interpreted

as follows: Variable $-y_{ij}$ says how much it costs to use edge (i, j) . Variable $\bar{y}_k$ says how much is saved by finding an alternative route from s_k to t_k .

The dual variables for the above model (11) are $y_{BD} = -13$ while all other are $y_{ij} = 0$. Furthermore, $\bar{y}_1 = 23$, $\bar{y}_2 = 60$, $\bar{y}_3 = 92$.

Question 20 (3 point) Write up an equation stating how the reduced cost of a route r can be calculated. Next, calculate the reduced cost of all routes from Question 14 given the above dual variables. Indicate the route $r \in R_1 \cup \dots \cup R_m$, having the most negative reduced cost. ■

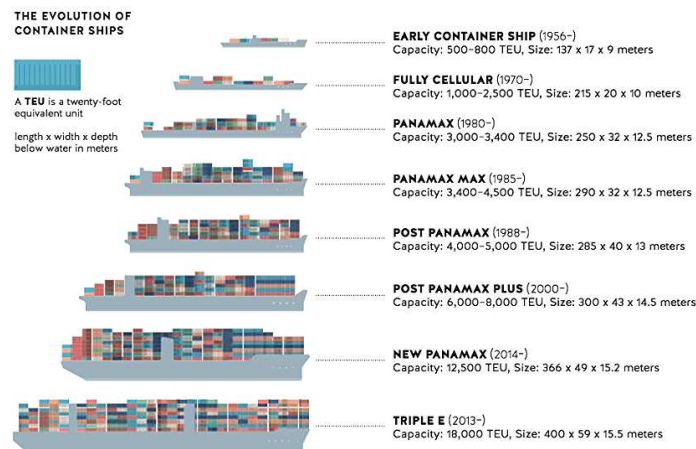
Since there can be an exponential number of routes for a given problem, calculating the reduced cost of all routes may be very time consuming, so we will use a more efficient approach to find the route with most negative reduced cost.

Question 21 (4 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Describe how the edge costs can be calculated, and discuss whether they can become negative. Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem as well as the reduced cost. ■

Question 22 (1 point) Extend the restricted master problem (11) with the new route. Solve the new problem to LP-optimality with CPLEX and report the dual variables. ■

Question 23 (4 point) Repeat the process of adding new promising routes to the restricted master model until no more routes with negative reduced cost can be added to the model. For each iteration, find the route with most negative reduced cost, using a shortest path algorithm. In each iteration write up the restricted master problem, and report the primal and dual variables as well as the objective. For each shortest-path problem solved, describe the graph, and report the solution. ■

Question 24 (2 point) Discuss how many routes were added in the column generation approach and compare it to the total number of routes. ■



Tips and tricks

Integer variables

In CPLEX you declare integer variables with the command `general`. If you use `integer` to declare variables then they become binary in CPLEX.

Bounds

Avoid using upper bounds on variables in CPLEX. Bounds are handled implicitly in Simplex, and hence the dual variables will *not* be reported. This may lead to wrong calculations of the reduced costs.

Upper bounds on variables can be formulated as constraints. However, it is easier to avoid all bounds on the x -variables. E.g. for binary variables x it is not necessary to have constraints $x \leq 1$, since x will never be larger than 1 in an optimal solution. The bound $x \geq 0$ is implicitly assumed for all LP-problems.

Checking your calculations in column generation

Assume that all columns are known in advance, but only a subset of the columns are used in the restricted master problem. Add a constraint

$$x_r^k \leq 0 \quad (12)$$

to all columns which are not used. This will “inactivate” the columns, and the same input file can be used for several iterations with only small modifications. The dual variable corresponding to (12) is actually the reduced cost of the column and can be used to check your calculations.

Output from CPLEX

If you want to use the output from CPLEX in your report, you can mark the text on the screen, and copy it into your document. Alternatively, you can use the log file from CPLEX. It can be found in the same directory as your input file. CPLEX writes all output to the log file.

